Métodos Matemáticos de la Física II

Juan I. Perotti*
Instituto de Física Enrique Gaviola (IFEG-CONICET),
Ciudad Universitaria, 5000 Córdoba, Argentina and
Facultad de Matemática, Astronomía, Física y Computación,
Universidad Nacional de Córdoba, Ciudad Universitaria, 5000 Córdoba, Argentina
(Dated: May 19, 2026)

Bla bla bla... resumen...

I. METRIC SPACE

A set X is called a *metric space* if it is equiped with a notion of *metric distance*, i.e. a function $d : X \times X \to \mathbb{R}$ such that, for any three elements x, y and z in X, which are called *points*, it holds:

1. $d(x, y) \geq 0$, i.e. it is *non-negative*.
2. $d(x, y) = 0$ implies $x = y$, i.e. it is *positive definite*.
3. $d(x, y) = d(y, x)$, i.e. it is *symmetric*.
4. $d(x, z) \leq d(x, y) + d(y, z)$, i.e. it satisfies the *triangular inequality*.

Example: The set $\mathbb{R}^n$ with the euclidean distance $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$ is a metric space.
Remark: There are metric spaces that are not finite-dimensional.

A. Open ball

The set

$$B_r(x) = \{y \in X : d(x, y) < r\}$$

is called the *open ball* of X centered at x and of radious $r > 0$.

B. Open set

A subset U of X is *open* if for every $x \in U$ there exists an open ball $B_r(x) \subset U$.

C. Complement set

Consider two subsets Y and Z of X such that $Y \subset Z$. The *complement* $Y \backslash Z$ of Y in Z is the subset of X given by

$$Y \backslash Z := \{y \in Y : y \notin Z\}.$$

D. Closed set

A subset F of X is *closed* if its complement $F \backslash X$ is open.

E. Limit or accumulation point

A point x is a *limit point* (or *accumulation point*) of a subset A of X if, for every $r > 0$, it holds that

$$(B_r(x) \backslash \{x\}) \cap A \neq \emptyset.$$

Informally, this means that there are points in A that are different but arbitrarily close to x.

Example: The points 0 and 1 are limit points of the open interval $(0, 1)$.

F. Closure set

The *closure* $\overline{A}$ of a subset A of $\mathbb{R}^n$ is defined by

$$\overline{A} = A \cup \{\text{all limiting points of } A\}.$$

Examples: the closed interval $[0, 1]$ is the closure of itself and the subsets $(0, 1)$, $[0, 1)$, $(0, 1]$ and $(0, 1/2) \cup (1/2, 1)$.

G. Bounded set

A subset A of X is *bounded* if there exists a finite $r > 0$ such that $A \subset B_r(0)$.

H. Compact set

When X is a finite-dimensional metric space, a subset K of X is called *compact* if it is closed and bounded.

Remark: The notion of compactness works differently in infinite-dimensional metric spaces. See appendix A.

I. Convergence

We say that a sequence $\{x_j \in X : j \in \mathbb{N}\}$ *converges* to a value $x \in X$ if, for each $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $d(x_j, x) < \epsilon$ for every $j \geq N$. It can be easily shown that, when the sequence is convergent, then the value of x is unique.

The convergence of the sequence is abbreviately denoted by $x_j \to x$ as $j \to \infty$.

J. Cauchy sequence

A sequence $\{x_j \in X : j \in \mathbb{N}\}$ is a *Cauchy sequence* if, for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $d(x_i, x_j) < \epsilon$ for every $i, j \geq N$.

K. Complete metric space

A metric space X equipped with a metric d is called *complete* if every Cauchy sequence in it converges to an element in it. More formally, if $\{x_j \in X : j \in \mathbb{N}\}$ is a Cauchy sequence, then there exists an $x \in X$ such that, for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $d(x_j, x) < \epsilon$ for every $j \geq N$.

Example: Consider the sequence given by

$$x_j = \sum_{i=0}^{j} \frac{1}{i!}.$$

Clearly, $x_j \to e$ as $j \to \infty$, i.e. it converges to Euler's number. This is a Cauchy sequence in $\mathbb{Q}$ but it does not converge to a value in $\mathbb{Q}$ because e is irrational. It converges to a value in $\mathbb{R}$. Therefore, $\mathbb{Q}$ is not a complete space.

II. NORMED VECTOR SPACE

A *normed vector space* V over a field $\mathbb{F}$ is a vector space equipped with a norm, which is a mapping $V \ni v \to |v| \in \mathbb{R}$ that, for any $v, w \in V$ and $\alpha \in \mathbb{F}$, it satisfies:

1. triangular inequality, $|v + w| \leq |v| + |w|$,
2. absolute homogeneity, $|\alpha v| = |\alpha||v|$, and
3. positive definiteness, $|v| = 0$ implies $v = 0$.

In particular, a *semi-norm* is a norm without property 3.

A. A normed vector space is a metric space

A normed vector space is a metric space because a norm defines a metric distance

$$d(v, w) = |v - w|.$$

To see this, it is enough to check that it actually satisfies the triangle inequality because the other properties trivially follow. Namely,

$$d(v, w) = |v - w| = |v - u + u - w| \leq |v - u| + |u - w| = d(v, u) + d(u, w)$$

holds for any $v, u, w \in V$ due to property 1 of the norm.

B. Banach space

A complete normed vector space is called a *Banach space*. In other words, as we already saw, a normed vector space is a metric space. If additionally, such metric space is complete, meaning that all Cauchy sequences in it converge to vectors in it, then it is called a Banach space.

C. Continuous function

Consider two metric spaces X and Y. A function $f : X \to Y$ is continuous at $x \in X$ if, for every convergent sequence $x_j \to x$ it holds that $f(x_j) \to f(x)$.

III. HILBER SPACE

Appendix A: Compactness in infinite dimension

The compactness of a subset K of a metric space X only makes sense if X is finite-dimensional? No. Compactness is a fundamental topological notion and makes perfect sense in *any* metric space, including infinite-dimensional ones. For a subset $K \subset X$ of a metric space (X, d), compactness means:

> Every open cover of K admits a finite subcover.

In metric spaces, this is equivalent to:

> Every sequence in K has a convergent subsequence whose limit lies in K.

This definition does not depend on dimension at all. What *does* change drastically between finite- and infinite-dimensional spaces is **which sets are compact**.

1. Finite-dimensional case

In $\mathbb{R}^n$, the Heine–Borel theorem says:

$$K \subset \mathbb{R}^n \text{ is compact} \iff K \text{ is closed and bounded.}$$

This is a very special property of finite-dimensional normed spaces.

2. Infinite-dimensional case

In infinite-dimensional normed spaces, closed and bounded sets are *usually not compact.* For example, consider the Hilbert space

$$\ell^2 = \left\{ x = (x_1, x_2, \dots) : \sum_{n=1}^{\infty} |x_n|^2 < \infty \right\}.$$

Its closed unit ball

$$B = \{x \in \ell^2 : |x| \leq 1\}$$

is closed and bounded, but **not compact**. A standard proof uses the canonical basis vectors

$$e_1 = (1, 0, 0, \dots), \quad e_2 = (0, 1, 0, \dots), \quad \dots$$

which satisfy

$$|e_n - e_m| = \sqrt{2} \quad (n \neq m).$$

Hence no subsequence can be Cauchy, so no subsequence converges. Therefore B is not compact.

3. What replaces Heine–Borel?

In metric spaces, compactness is closely related to **total boundedness**.
For metric spaces:

$$K \text{ compact} \iff K \text{ complete and totally bounded.}$$

Total boundedness is stronger than ordinary boundedness. A set is totally bounded if:

For every $\varepsilon > 0$, it can be covered by finitely many balls of radius ε.

In finite dimensions, boundedness already implies total boundedness. In infinite dimensions, it generally does not.

4. Intuition

Infinite-dimensional balls contain "too many independent directions." The sequence e_n in ℓ^2 keeps jumping into new orthogonal directions and never accumulates anywhere. That phenomenon cannot happen inside bounded subsets of finite-dimensional spaces. So compactness absolutely makes sense in infinite-dimensional spaces — it simply becomes much more restrictive.

* juan.perotti@unc.edu.ar